

WASP-1b

R-1370

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-1_151011 · created 2026-08-01T00:42:56+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2457306.8707 +/- 0.0071 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.13 +/- 0.039
Transit depth (Rp/Rs)^2	1.68 +/- 1.02 [%]
Orbital Inclination (inc)	85.02 +/- 1.42
Airmass coefficient 1 (a1)	1.086 +/- 0.058
Airmass coefficient 2 (a2)	-0.078 +/- 0.075
Scatter in the residuals of the lightcurve fit is	5.16 %
Best Comparison Star	#1 - [617, 33]
Optimal Aperture	2.94
Optimal Annulus	11.64
Transit Duration (day)	0.141 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.13 ± 0.039 vs 0.1036 (deviation 0.68σ)
Duration fit vs published	0.141 d vs 0.1567 d (deviation 1.42σ)
Mid-time O–C	-17.5 min
Depth SNR	3.3
Residual scatter	5.16 %

Light curve

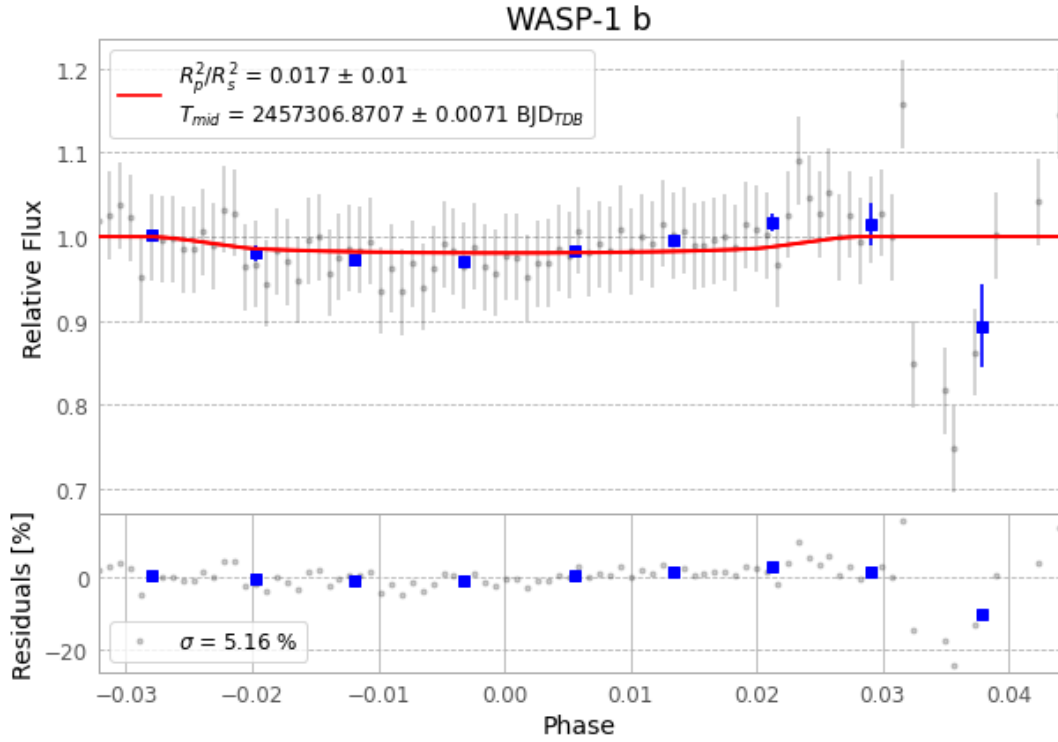


Figure 1 — FinalLightCurve_WASP-1 b_2015-10-10.png (SHA-256 b966d3b4dba1cbbd...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [188, 269], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 26c5bf0c11926c6a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

103 FITS frames (68 MB), WASP-1151011064812.FITS → WASP-1151011115412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-1-151011.log (da3905cfd0b9...), FinalLightCurve_WASP-1 b_2015-10-10.png (b966d3b4dba1...), FinalLightCurve_WASP-1 b_2015-10-10.csv (351223569267...)

Compute resources

Wall time 260.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 00:42Z.